

22GE003 & BASICS OF ELECTRICAL
ENGINEERING

1. * Electrical field exerts force along or opposite to the motion of the proton.
* Electric field does work on the particle, so its kinetic energy and speed can change.
* Magnetic field acts perpendicular to motion, changing only direction, not the speed.
2. * using $F = qvB$, if the magnetic field B is doubled, the magnetic force F also doubles because force is directly proportional to the magnetic field.
3. * velocity selector allows only particles with a specific velocity to pass straight.
* Particles with other velocity are deflected and removed.
* This helps achieve accurate measurement and separation in instruments like mass spectrometer.

4. Sol Given : $E = 2000 \text{ N/C}$, $B = 0.02 \text{ T}$

$$\text{Velocity } (v) = \frac{E}{B} = \frac{2000}{0.02} = 100000 \text{ m/s}$$

$\therefore$ velocity of undeflected particles $= 1 \times 10^5 \text{ m/s}$

5. * using $F = qvB \sin \theta$
* given: velocity is parallel to magnetic field. $\therefore \theta = 0^\circ$
* since, $\sin 0^\circ = 0$, magnetic force become zero.
* $\therefore$ no magnetic force acts on the particle and it continues in a straight path.

6. Sol

Given: $q = 4 \times 10^{-6} \text{ C}$, $E = 2500 \text{ N/C}$

$$F = qE$$

$$= (4 \times 10^{-6})(2500)$$

$$F = 0.01 \text{ N} \text{ (Electric force)}$$

7. * Electric force and magnetic force act in opposite directions on the particle
* Only at a particular velocity, both forces become equal.
* Hence, net force becomes zero and the particles and the particles pass undeflected.

8. * Using $v = \frac{E}{B}$, If the magnetic field B is doubled, the selected velocity v becomes half because velocity is inversely proportional to the magnetic field.

9. Given: $q = 4 \times 10^{-6} \text{ C}$, $E = 5 \times 10^3 \text{ N/C}$

$$F = qE = 4 \times 10^{-6} \times 5 \times 10^3$$

$$F = 2 \times 10^{-2} \Rightarrow F = 0.02 \text{ N}$$

* Using $F = qE$

- Force is directly proportional to the charge
- If the charges increase, the electric field also increases.
- A larger force causes greater acceleration of the particle

10. Given: $B = 0.40 \text{ T}$, $I = 5 \text{ A}$, $L = 0.30 \text{ m}$

$$F = BIL = 0.40 \times 5 \times 0.30$$

$$F = 0.6 \text{ N} \text{ Force acting on the conductor.}$$

- * Magnetic force is directly proportional to the magnetic field strength (B)
- * Increasing the magnetic field increases the force on the conductor
- * Greater force produces stronger mechanical motion.
- * This principle is used in electric motors to create rotation.

11. Given: $E = 3 \times 10^4 \text{ NC}^{-1}$, $q = 1.6 \times 10^{-19} \text{ C}$

$$F = qE$$

$$F = 3 \times 10^4 \times 1.6 \times 10^{-19}$$

$$F = 4.8 \times 10^{-15} \text{ N}$$

- * The electric field accelerates the electron. Greater electric field gives greater force
- * This principle is used in particle accelerators.

12. Using $F = BIL$,

- * A current-carrying conductor produces moving charges inside the magnetic field
- * The magnetic field exerts force on these moving charges.
- * This creates a force on the conductor perpendicular to both current and magnetic field
- * In an electric motor, forces on opposite sides of the coil produce ~~motion~~ rotation.
- * Thus, electrical energy is converted into mechanical motion.

13. given: $q = 1.6 \times 10^{-19} \text{ C}$, $v = 3 \times 10^6 \text{ m/s}$, $B = 0.03 \text{ T}$

$$F = qvB$$

$$F = 1.6 \times 10^{-19} \times 3 \times 10^6 \times 0.03$$

$$F = 1.44 \times 10^{-14} \text{ N}$$

- * Force is directly proportional to magnetic field strength B .
- * Increasing B increases the magnetic force
- * Higher force causes tighter circular motion.
- * This principle is used in cyclotrons to control particle path.

14.

given: $I_1 = 8 \text{ A}$, $I_2 = 6 \text{ A}$, $L = 0.50 \text{ m}$, $d = 0.05 \text{ m}$, $\mu_0 = 4\pi \times 10^{-7} \text{ Tm/A}$

$$F = \frac{\mu_0 I_1 I_2 L}{2\pi d}$$

$$F = \frac{4\pi \times 10^{-7} \times 8 \times 6 \times 0.50}{2\pi}$$

$$F = \frac{4 \times 10^{-7} \times 24 \times 0.50}{1}$$

$$F = \frac{4.8 \times 10^{-6}}{0.10}$$

$$F = 4.8 \times 10^{-5} \text{ N}$$

- * The conductors attract each other because same direction current attracts each other.

15. * Only the velocity component perpendicular to the magnetic field experiences force.
- * This force is perpendicular to motion, causing circular movement.
 - * The parallel velocity remains unchanged.
 - * Combination of circular and straight motion forms a helix structure.
 - * Hence, the particle follows a helical path in the magnetic field.
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16. * A current in each conductor produces a magnetic field around it.
- * This magnetic field exerts force on the moving charges in the other conductor.
 - * The interaction of these fields create a force between the wires.
 - * Same direction current produce attraction, opposite direction produce repulsion.
 - * This is due to the direction of magnetic force on charges in each wire.
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17. * The transmitter produces time-varying electric and magnetic field, forming electromagnetic waves.
- * These waves carry energy through space without any physical connection.
 - * When the waves reaches the detector, its electric field induces a current, allowing the detector to receive and record the transmitted energy.

18. * Solar energy is transferred by radiation.
* The sun emits electromagnetic waves which travel through empty space and carry energy to Earth without needing a medium.

19. * When the metal plate moves in a magnetic field, the magnetic flux through it changes.
* This induces eddy currents in the plate according to electromagnetic induction.
* These eddy currents produce a magnetic field that opposes the motion (Lenz's law), causing resistance to movement.

20. * A changing magnetic field induces circulating currents in a conductor.
* These currents flow in closed loops within the metal.
* They are called eddy current.

21. * The transmitter produces oscillating electric and magnetic fields, forming electromagnetic waves.
* These waves carry energy through space without any physical medium.
* The receiver absorbs the wave and converts it into electrical signal energy.

22. * The phenomenon responsible for is electromagnetic wave propagation (radiation).

* Energy is carried through space by electromagnetic waves without any physical medium.

23. * Rotation in a magnetic field induces eddy currents in the copper disk.
* According to Lenz's law, These currents oppose the motion.
* This creates a retarding force (magnetic damping), slowing down the disc.

24. * The phenomenon is electromagnetic induction.
* The changing magnetic field induces eddy currents in the aluminium plate, which oppose the motion.

25. * current in an inductor creates a magnetic field around the coil.
* The magnetic field stores energy in the form of magnetic potential energy
* The energy stored is proportional to the square of the current ($E \propto I^2$)
* When current changes, the magnetic field collapses and releases the stored energy back into the circuit.
* Thus, inductors store and regulate energy using magnetic fields, helping smooth current changes.

26. * Self inductance is the property of a coil by which a change in current induces an emf in the same coil.

* The induced emf is due to the change in magnetic flux linked with the coil.

* The induced emf opposes the change in current (Lenz's law)

* It depends on the geometry of the coil and number of turns.

* Thus, self inductance help oppose sudden current changes and stabilize circuits.

27. Same as 25th question

28. Same as 26th question

29. * In a generator, a coil rotates inside a magnetic field

* Rotation causes continuous change in magnetic flux linked with coil.

* This changing magnetic field induces an emf in the coil (Faraday's law)

* The induced emf drives electric current through

the external circuit.

* Thus, electromagnetic induction converts mechanical energy into electrical energy

30. * Electromagnetic radiation carries both energy and momentum.

* When radiation strikes a particle, momentum is transferred to it.

* This transfer exerts a small force called radiation pressure.

* The particle moves in the direction of the ~~magnetic field~~ electromagnetic wave.

* Thus, electromagnetic waves can transfer momentum and exert force on matter.

31. * Alternating current in the primary coil produces a changing magnetic field.

* This changing field links with the nearby secondary coil.

* The magnetic flux through the secondary coil continuously changes.

* According to Faraday's law, this change induces an emf in the secondary coil.

* Thus, electrical energy is transferred between ~~the~~ coils by electromagnetic induction without ~~direction~~ contact.

32. Same as 30th qn.